Zhijie Wang

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EDUCATION

University of Wisconsin - Madison Madison, US
Ph.D. in *Civil and Environmental Engineering* (in progress) GPA 3.942/4.00 Jan 2024 – May 2028
Minor in *Computer Science* with AI-focused coursework

Tongji University Shanghai, China
Master of Science in *Environmental Engineering*, GPA 86.56/100 Sep 2020 – Jun 2023
Erasmus+ programme of EU funded Master Program on bio-based circular economy

University of Florence Florence, Italy
Semester exchange, Co-op intern at NOVAMONT S.p.A Sep 2022 – Jan 2023

Shandong University Tsingtao, China
Bachelor of Science in *Environmental Engineering*, GPA 4.55/5.00 Sep 2016 – Jun 2020
Summer camp in Sungkyunkwan University, China-Korea-Japan cooperative project for nurturing leading infrastructure engineers

RESEARCH EXPERIENCE

Research Assistant
Qin Lab, Department of Civil and Environmental Engineering, UW-Madison Jan 2024 – Present
Advisor: Dr. Mohan Qin

Fate transformation of microplastics under pressure-driven membrane filtration

- Study the plastic fragmentation during pressurized filtration system
- Develop AI-assisted automated low-micrometer microplastic detection workflow integrating image segmentation and spectrum classification
- Design Raman spectrum processing algorithm for low SNR data features extraction and classification

Resource recovery from digested high-strength waste using pressure-driven membrane

- Model the nutrient mass transfer across an intrinsic charged membrane
- Develop theory framework describing the transmembrane transport of amphoteric ions (NH_3 and NH_4^+)

Institute of Solid Waste Treatment and Reclamation, Tongji University Sep 2020 – Jun 2023
Advisor: Dr. Fan Lü

Municipal solid waste reclamation towards value-added organic acids production and recovery

Topic of Master Honors Thesis

- Designed an anaerobic membrane bioreactor coupled with electrodialysis unit to produce and recover organic chemicals, primarily short and medium chain carboxylic acids
- Investigated transport of salts, nutrients, and organics during electrodialysis and membrane fouling mechanisms

Degradability of bio-based plastic during compost and anaerobic digestion with food waste

- Designed compost and anaerobic digestion experiments where bioplastic degraded with food waste
- Determined the physical and chemical changes of bioplastic during degradation

RESEARCH EXPERIENCE (CONTINUED)

Undergraduate Researcher

School of Environmental Science and Engineering, Shandong University Sep 2018 – Jun 2020

Biochar-mediated enhanced anaerobic methanogenesis and PAH degradation

Topic of Undergraduate Honors Thesis, funded by National College Student Innovation Supporting Program

- Domesticated anaerobic sludge with high PAH resistance and investigate its capability of degrading PAH
- Enhanced the performance of methanogens by introducing biochar into anaerobic system

TEACHING EXPERIENCE

Teaching Assistant Jan 2025 – May 2025

Undergraduate course: *Environmental Engineering*, UW-Madison

- Attended lectures and discuss sessions to provide support to faculty and students while teaching and completing assignments; Gave lectures during faculty absence
- Supported student learning outside the classroom via facilitation of office hours (weekly) and Q&A/review sessions
- Developed written solutions and grading rubrics for problem sets, quizzes, and exams
- Graded exams and provided input, feedback, and support for the primary course instructors

Teaching Assistant

Undergraduate course: *Statics*, Vanderbilt University Aug 2023 – Dec 2023

- Graded and provided educational feedback and commentary on homework assignments and exams
- Supported students learning via weekly office hour to provide additional tutoring and academic support

Teaching Assistant

Undergraduate course: *Solid Waste Management*, Tongji University Sep 2020 – Jan 2021

- Prepared lecture materials, handouts, conceptual illustrations, and slides to support faculty teaching
- Graded and provided educational feedback and commentary on homework assignments and exams

INDUSTRY EXPERIENCE

NOVAMONT S.p.A. Novara, Italy

Internship at the Headquarter Oct 2022 – Dec 2022

Department of Strategy and Agriculture Public Affairs

- Collected and reported the Chinese and EU legislative framework on agricultural policy and biodegradable and compostable materials
- Created preliminary engagements with relevant stakeholders for the above-mentioned topics
- Followed the relevant conferences, workshops, and exhibitions on agriculture and plasticulture
- Organized, monitored, and prepared reports on the agronomical experimental fields organized using biodegradable and compostable solutions from Novamont

AWARDS & HONORS

Dean's Graduate Fellowship of Vanderbilt University — Aug 2023

Awarded to new graduate students joining the university to recognize academic merit and the potential future contributions

Outstanding Graduate of Shanghai Municipality — Jun 2023

Awarded to a member of the graduating class whose academic achievements, leadership, and community services stand out among all the graduating student in Shanghai, including students from other top universities in Shanghai

Chinese Ministry of Education's National Graduate Scholarship — Oct 2022

Awarded to graduate students who have the best academic records (top 2%) among peers

Representative of National Scholarship recipient reported by People's Daily — May 2023

People's Daily is the largest newspaper in China, and each top university can nominate only one National Fellowship awardee who has the most outstanding achievements per year for newspaper coverage

Recipient of the Chongyue's Environmental Education Scholarship — Oct 2021

Awarded to graduate students in Environment-related major who exemplify excellent research abilities

Outstanding Graduate of Shandong University — Jun 2020

Awarded to a member of the graduating class whose academic achievements, leadership, and community services stand out among all the graduating student in Shandong University

Recipient of the Academic Merit Fellowship for three years, Shandong University — 2017-2020

Awarded to undergraduate students who have the impressive academic records (top 10%) among peers

PUBLICATIONS

[1] **Wang Z**, Yang K, Mahajan S, et al. Understanding and modeling the partitioning and transport of NH_3/NH_4^+ across RO membrane. *Nature Comm.* 2026, https://doi.org/10.1038/s41467-026-71260-y

[2] **Wang Z**, Wu Z, Janssen E S, et al. Microplastics undergo fragmentation during pressure-driven membrane filtration. *Environ. Sci. Technol. Lett.* (In preparation)

[3] **Wang Z**, Wu Z, Janssen E S, et al. Role of membrane in micro/nano plastic research and its integration with artificial intelligence: advances, challenges, and prospects. *Anal. Chem.* (In preparation)

[4] **Wang Z**, He P, Huang Y, et al. Molecular behavior of dissolved organic matter in liquid digestate from biogenic waste during electrodialysis treatment: Reserve, transfer, or foul the membrane? *ACS ES&T Water*, 2023, 3, 7, 1803–1816. https://doi.org/10.1021/acsestwater.2c00302

[5] **Wang Z**, He P, Zhang H, et al. Desalination, nutrients recovery, or products extraction: Is electrodialysis an effective way to achieve high-value utilization of liquid digestate? *Chem. Eng. J.*, 2022, 446: 136996. https://doi.org/10.1016/j.cej.2022.136996

[6] **Wang Z***, Liu Y*, He P, et al. Short-chain carboxylate continuous production from food waste with in-situ extraction: Novel design to reduce chemical input for pH control and alleviate membrane fouling. *J. Environ. Chem. Eng.,* 2024, 12(3): 112554. https://doi.org/10.1016/j.jece.2024.112554 (*equal contribution)

PUBLICATIONS (CONTINUED)

[7] Peng W, **Wang Z**, Shu Y, et al. Fate of a biobased polymer via high-solid anaerobic co-digestion with food waste and following aerobic treatment: Insights on changes of polymer physicochemical properties and the role of microbial and fungal communities. *Bioresour. Technol.*, 2022, 343: 126079. https://doi.org/10.1016/j.biortech.2021.126079

[8] Lü F, **Wang Z**, Zhang H, et al. Anaerobic digestion of organic waste: Recovery of value-added and inhibitory compounds from liquid fraction of digestate. *Bioresour. Technol.*, 2021, 333: 125196. https://doi.org/10.1016/j.biortech.2021.125196

[9] **Wang Z**, He P, Liu Y, et al. Maximizing Hydrolysis and Acidification of OFMSW Post-Transportation: Toward Easily Degradable Dissolved Carbon Production. *Waste and Biomass Valorization*, 2024, 15: 3269-3280. https://doi.org/10.1007/s12649-023-02398-9

[10] **Wang Z**, He P, et al. Analysis on domestic and foreign standards and policies about the land application of anaerobic digestate. *Environmental Sanitation Engineering* (In Chinese), 2022, 30(1): 17-27. https://doi.org/10.19841/j.cnki.hjwsgc.2022.01.003

PATENTS

[1] A device for in-situ organic acids recovery from the organic fraction of municipal solid waste. China Invention Patent. ZL 2021 1 0885246.1

[2] Design of modularized volatile fatty acid production reactor and workflow for products recovery from organic-rich waste stream. China Invention Patent. (Under review)

CONFERENCE PRESENTATIONS

Wang Z and Qin M, Fate transformation of low-micrometer microplastics during pressure-driven membrane filtration. IWA LET 2026, Jun 1st – Jun 5th, Rice University, Houston TX, US. *Accepted for oral presentation.*

Wang Z and Qin M, Enhance the recovery of NH_3/NH_4^+ using RO membrane. AEESP 2025, May 19th – 22nd, Duke University, Durham NC, US. *Oral presentation.*

Wang Z and Qin M, Understanding the transport of NH_3/NH_4^+ across RO membrane. ACS Spring 2025, March 23rd – 27th, San Diego CA, US. *Oral presentation.*

Wang Z, He P, Zhang H, et al. Migrating patterns of inorganic ions, nutrients, and organics in liquid digestate during electrodialysis. The 11th National Chemical and Environmental Conference, 25th – 29th July 2022, Harbin Institute of Technology, Harbin, China. *Poster presentation*.

Wang Z, He P, Zhang H, et al. Recovery of value-added and inhibitory compounds from liquid fraction of digestate. The Ending Conference of EU Erasmus+ programme funded Master Program (BBChina), 26th September 2021, Tongji University, Shanghai, China. *Oral presentation.*

SKILLS & INTERESTS

- Language: Chinese – Native, English – Fluent
- Software: Microsoft Office – Expert, Origin – Expert, AutoCAD – Proficient, Adobe Photoshop – Proficient, Adobe Premiere – Proficient, Adobe Illustrator – Competent
- Programming: Python – Expert, MATLAB – Expert, R – Proficient, Shell – Proficient, LaTeX – Proficient
- Lab Instruments: HPLC, Orbitrap LC-MS, GC, IC, ICP-OES, TOC, Raman microscopy, FTIR
- Interests: Photography, Snowboarding, Kayaking, Hiking